Long Nguyen Khac

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Education

International School, Vietnam National University, Hanoi — *Vietnam*
B.Eng. in Automation and Informatics — *Sep. 2021 – Dec. 2026*

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore (05/2025 – 05/2026).
- **Goal:** to graduate with Excellent distinction and the highest academic standing in the cohort.
- **Research interest:** focusing on adaptive control for AUVs integrated with wireless power charging system.

Research Experience

Water & Energy Research Lab, EEE, Nanyang Technological University — *Singapore*
Research Fellow — *Feb. 2026 – Mar. 2026*

- **Supervisor:** Asst. Prof. Yun Yang & PhD. Kaiyuan Wang
- Conducted research on relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configurations, power transfer efficiency, and system-level integration for docking-free underwater charging.

AISC Lab, AIM-HI, National Chung Cheng University — *Taiwan / Remote*
Research Intern — *May 2025 – Present*

- **Supervisor:** Prof. Her-Terng Yau & PhD.Hao-Yang Lin
- Applying machine learning models, including Random Forest, XGBoost, and Support Vector Regression, to predict cutting forces in CNC milling from Taguchi machining datasets.
- Developed data-driven modeling methods for tool wear & cutting force prediction in CNC milling using experimental datasets.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS — *Vietnam*
Student Researcher — *Sep. 2022 – Present*

- **Supervisor:** Dr. Le Xuan Hai
- Conducting research on adaptive control for AUVs, focusing on station keeping for planar wireless charging.
- Designed the mechanical model and physical prototype of a 3-DOF delta parallel robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Built a dataset from construction-site images and trained YOLOv5s-based models for PPEs detection.

International Experience

Nanyang Technological University — *Singapore*
Global Connect Fellow — *Feb. 2026 – Mar. 2026*

- **Program:** NTU Global Connect Fellowship
- Selected as 1 of 39 young scholars worldwide and 1 of 5 outstanding Vietnamese young scholars funded fully for the Global Connect Fellowship program at NTU, Singapore (∼2% acceptance rate from ∼2,700 applicants).

National University of Singapore — *Singapore*
Exchange Student — *Aug. 2025 – Dec. 2025*

- **Program:** DiscoverNUS
- Selected as one of 8 outstanding VNU students receiving a fully funded scholarship DiscoverNUS & the NUS ASEAN Master's Scholarship Programme.

National Chung Cheng University — *Taiwan*
Research Intern — *May 2025 – Jul. 2025*

- **Program:** Taiwan Experience Education Program (TEEP)
- Selected as one of 28 global students to intern at AIM-HI, CCU funded by Ministry of Education, Taiwan

Universitas Gadjah Mada & Kyushu University — *Indonesia*
VNU representative — *Feb. 2025*

- **Program:** ASEAN in Today's World
- Selected as one of four excellent VNU representatives to participate the program funded by Kyushu University, Japan.

Ngee Ann Polytechnic — *Singapore*
VNU Representative — *Feb. 2024*

- **Program:** Temasek Foundation Specialist's Community Action & Leadership Exchange
- Selected as one of 29 VNU students to receive a fully-funded scholarship for the program by Temasek Foundation.

Publications

1. **Long Nguyen Khac** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot.* International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
   - ***Main highlights:*** Adaptive fuzzy backstepping control for robust trajectory tracking of a 3-DOF delta parallel robot.
   - ***Presentation:*** Mechatronics and Automation section, The 3rd International Conference ICTA 2024.
2. **Long Nguyen Khac** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites.* Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
   - ***Main highlights:*** YOLOv5s-based PPE detection using six activation functions and a dataset of 2,632 construction-site images.
   - ***Presentation:*** National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

Honors & Awards

Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation — Mar. 2026
NUS ASEAN SPARK Fellow, NUS Enterprise Summer Programme in Entrepreneurship — Jul. 2026
Outstanding Young Face Award, Vietnam National University, International School, awarded in 2024 and 2025 — 2024 – 2025
Five Good Merits Award, Vietnam National University, Hanoi — Dec. 2024
Toshiba Scholarship, ToshibaInternational Foundation — Sep. 2024
Fighting Spirit Prize, The IEEE Southeast Asia Circuits and Systems Society Hackathon — Dec. 2022
Merit-based University Academic Scholarship, Six-semester recipient for excellent academic performance — 2021–2025

Skills

Research — Mathematical modeling, control systems design, autonomous systems (AUV, mobile robots), wireless power transfer, data-driven modeling
Programming — MATLAB, Python, Embedded C/C++, PLC programming
Software — MATLAB/Simulink, SolidWorks, Altium, Proteus, Visual Components, TIA Portal
AI — Data-driven modeling (SINDy, SVD), Machine learning (Random Forest, XGBoost), Computer vision (YOLO)
Languages — Vietnamese (Native), English (IELTS 6.5), French (Beginner), Chinese (Beginner)

References

Prof. Her-Terng Yau — Director of AIM-HI, National Chung Cheng University
Her-Terng Yau | htyau@ccu.edu.tw

Asst. Prof. Yun Yang — Assistant Professor, School of Electrical & Electronic Engineering, NTU
Yun Yang | yun.yang@ntu.edu.sg

Dr. Le Xuan Hai — Vice Dean, Faculty of Engineering and Technology, VNUIS
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